

Unified EDA

Updated formatting across all sections.

Added prevalence heatmaps for general health and demographics.

Correlation heatmaps retained; correlation tables exported separately as CSV and PNG.

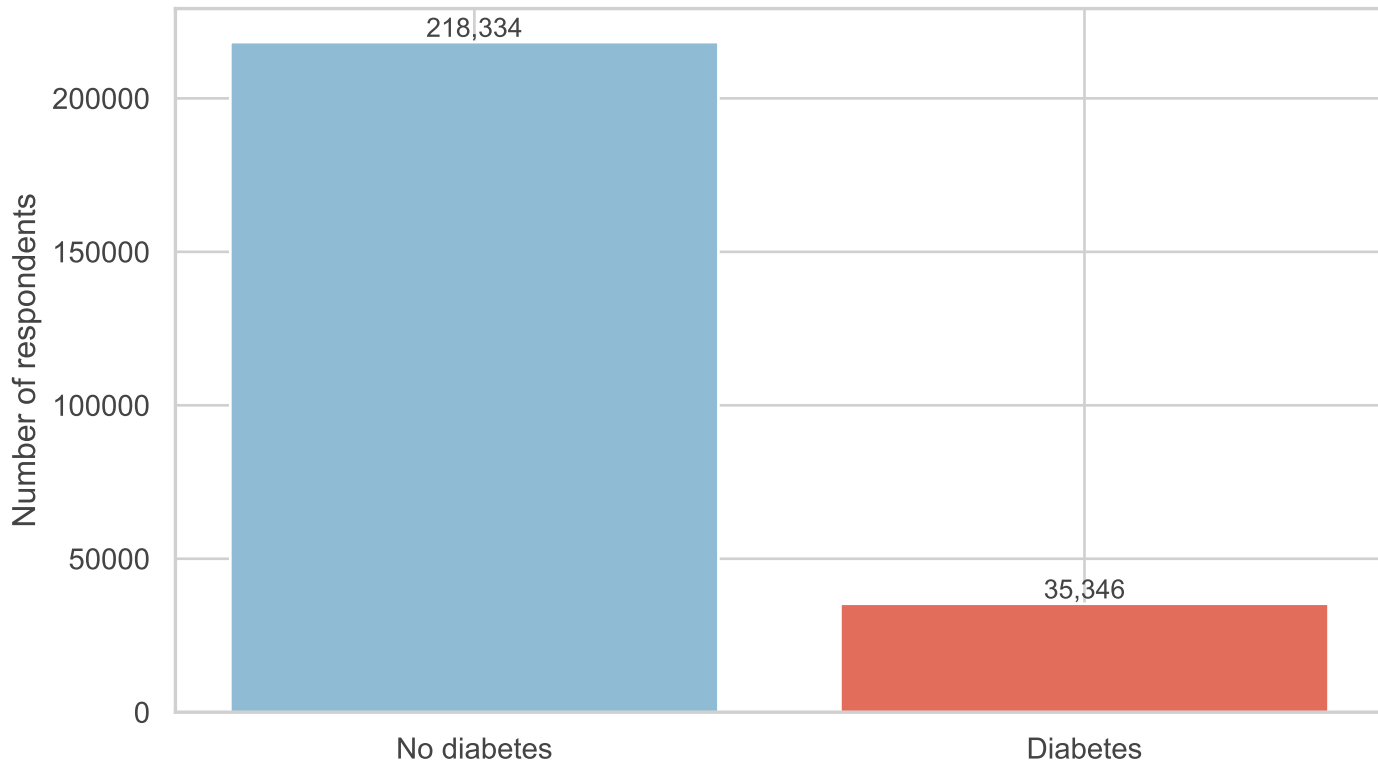
Dataset Overview

Metric	Value
Rows	253,680
Columns in all-variable dataset	25
Missing cells	0
No diabetes	218,334 (86.1%)
Diabetes	35,346 (13.9%)

Variable Dictionary

Variable	Type	Meaning
BMI	Continuous	Body Mass Index
HighBP	Binary	High blood pressure
HighChol	Binary	High cholesterol
CholCheck	Binary	Cholesterol check in past 5 years
Stroke	Binary	History of stroke
HeartDiseaseorAttack	Binary	History of coronary heart disease or heart attack
Smoker	Binary	Smoked at least 100 cigarettes in lifetime
PhysActivity	Binary	Physical activity in past 30 days, excluding job
GenHlth	Ordinal	Self-rated health from excellent to poor
PhysHlth	Count	Poor physical health days in past 30 days
MentHlth	Count	Poor mental health days in past 30 days
DiffWalk	Binary	Serious difficulty walking or climbing stairs
Age	Ordinal	Age group 18-24 through 80+
Sex	Binary	F/M indicator
Income	Ordinal	Income category from <10k to 75k+
Education	Ordinal	Education level from no school to college graduate
Diabetes_binary	Binary	No diabetes / diabetes

Outcome Distribution



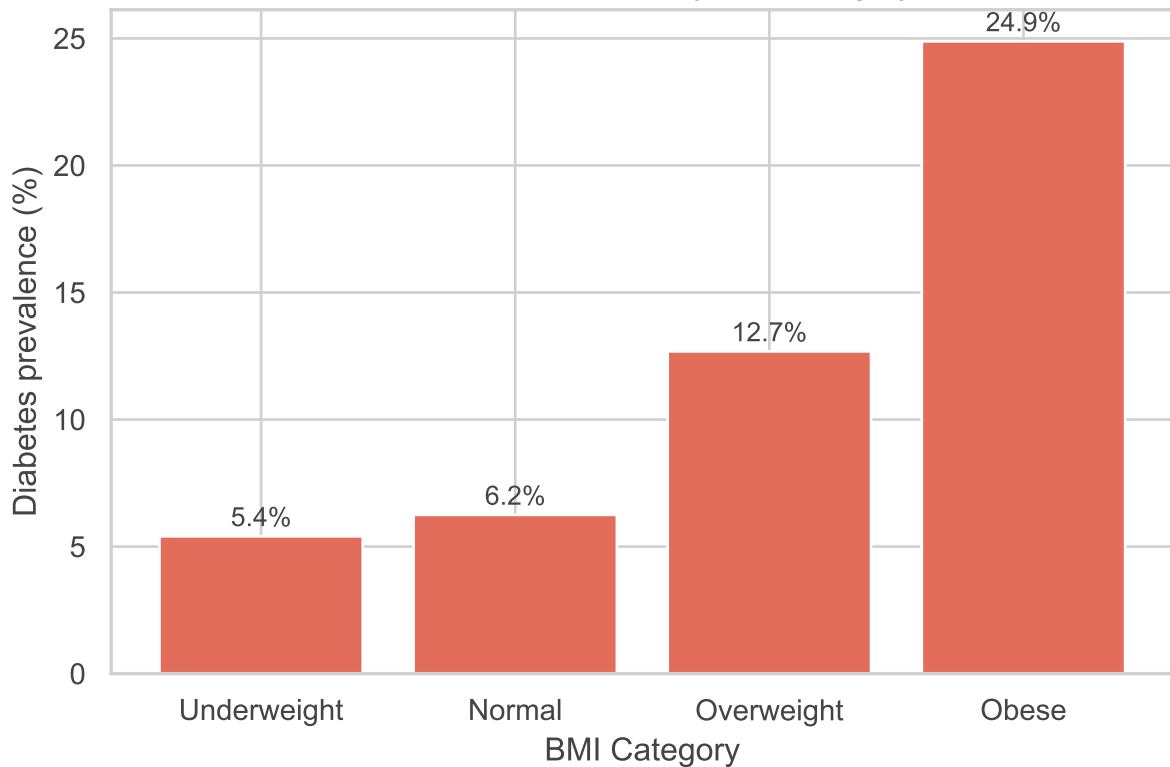
Cardiometabolic Baseline Comparison by Diabetes Status

Variable	No diabetes	Diabetes	Difference	SMD
BMI	27.806	31.944	4.138	0.604
HighBP	0.377	0.753	0.376	0.82
HighChol	0.384	0.67	0.286	0.597
CholCheck	0.958	0.993	0.035	0.231
Stroke	0.032	0.092	0.06	0.251
HeartDiseaseorAttack	0.073	0.223	0.15	0.431

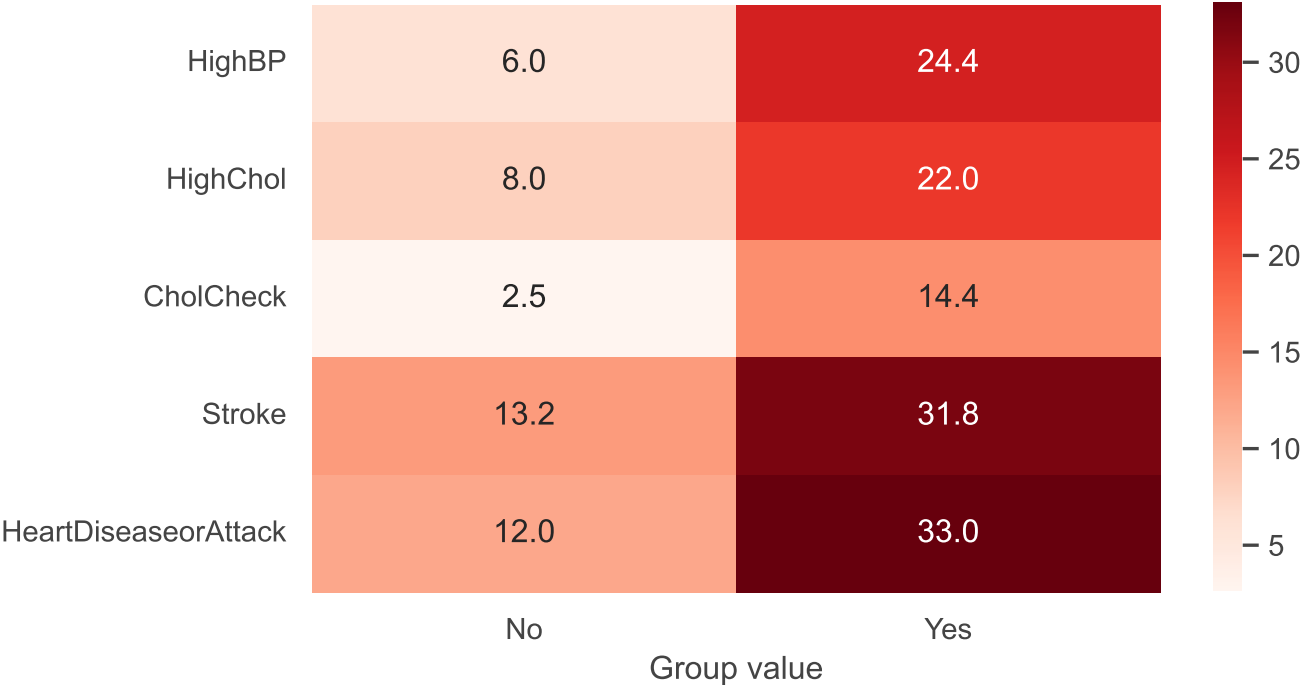
Cardiometabolic Diabetes Prevalence by Category

Variable	Category	Count	Diabetes prevalence
BMI_group	Underweight	3,127	5.4%
BMI_group	Normal	86,099	6.2%
BMI_group	Overweight	91,176	12.7%
BMI_group	Obese	73,278	24.9%
HighBP	No	144,851	6.0%
HighBP	Yes	108,829	24.4%
HighChol	No	146,089	8.0%
HighChol	Yes	107,591	22.0%
CholCheck	No	9,470	2.5%
CholCheck	Yes	244,210	14.4%
Stroke	No	243,388	13.2%
Stroke	Yes	10,292	31.8%
HeartDiseaseorAttack	No	229,787	12.0%
HeartDiseaseorAttack	Yes	23,893	33.0%

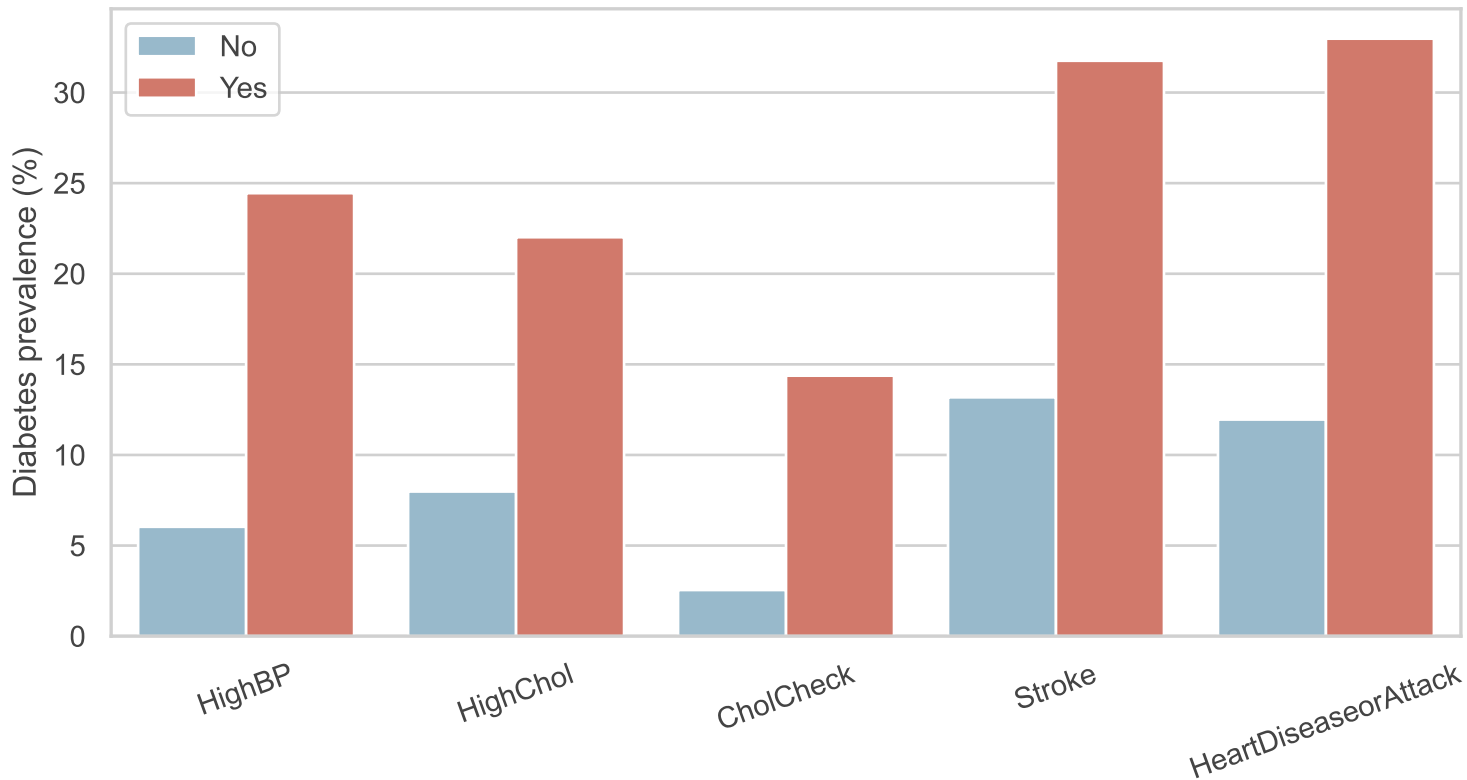
Diabetes Prevalence by BMI Category



Cardiometabolic Binary Variable Prevalence Heatmap



Cardiometabolic Binary Variable Prevalence



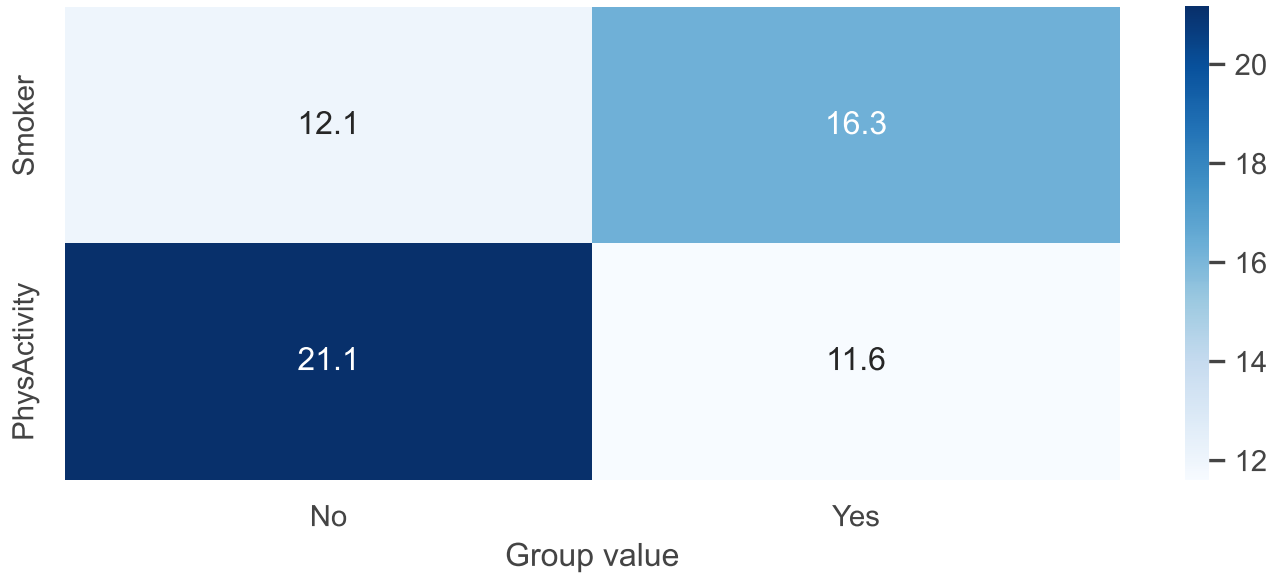
Lifestyle Baseline Comparison by Diabetes Status

Variable	No diabetes	Diabetes	Difference	SMD
Smoker	0.431	0.518	0.087	0.175
PhysActivity	0.777	0.631	-0.146	-0.325

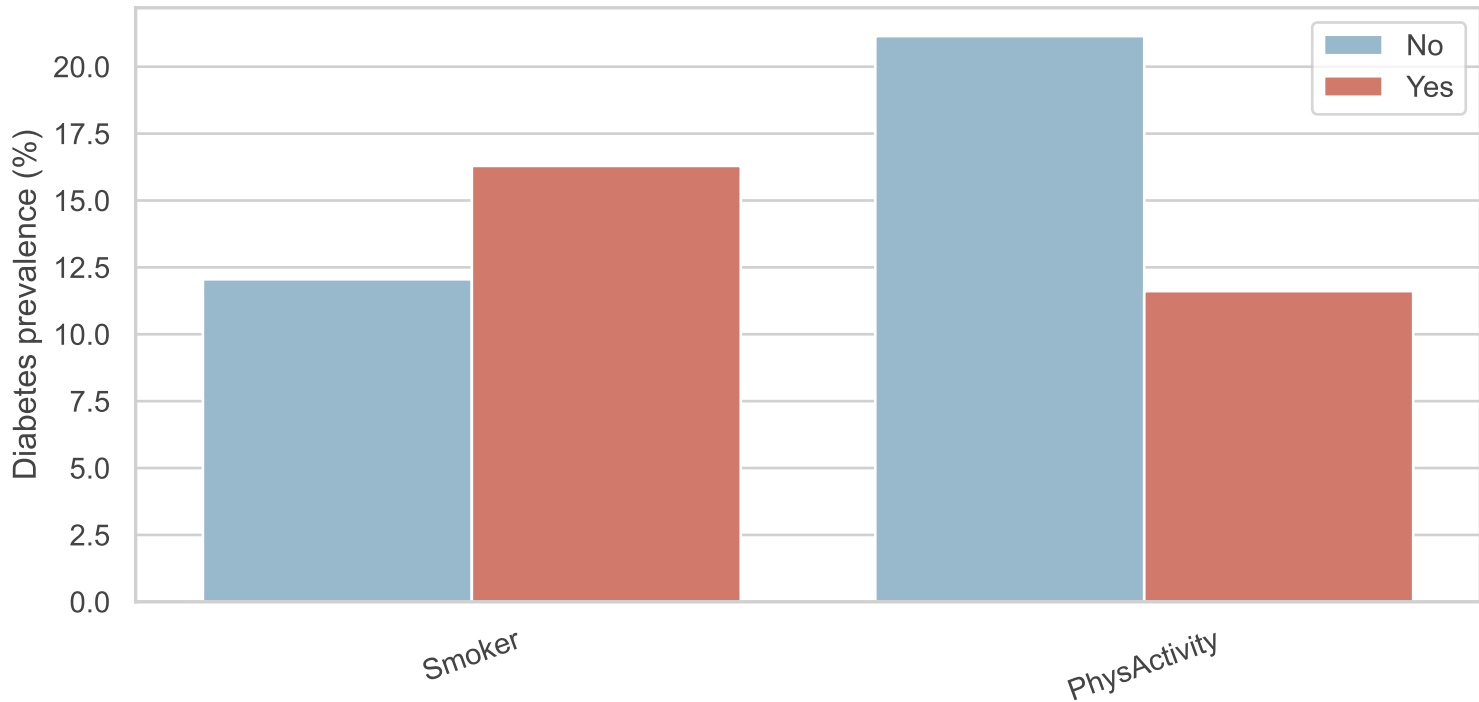
Lifestyle Diabetes Prevalence by Category

Variable	Category	Count	Diabetes prevalence
Smoker	No	141,257	12.1%
Smoker	Yes	112,423	16.3%
PhysActivity	No	61,760	21.1%
PhysActivity	Yes	191,920	11.6%

Lifestyle Variable Prevalence Heatmap



Lifestyle Variable Prevalence



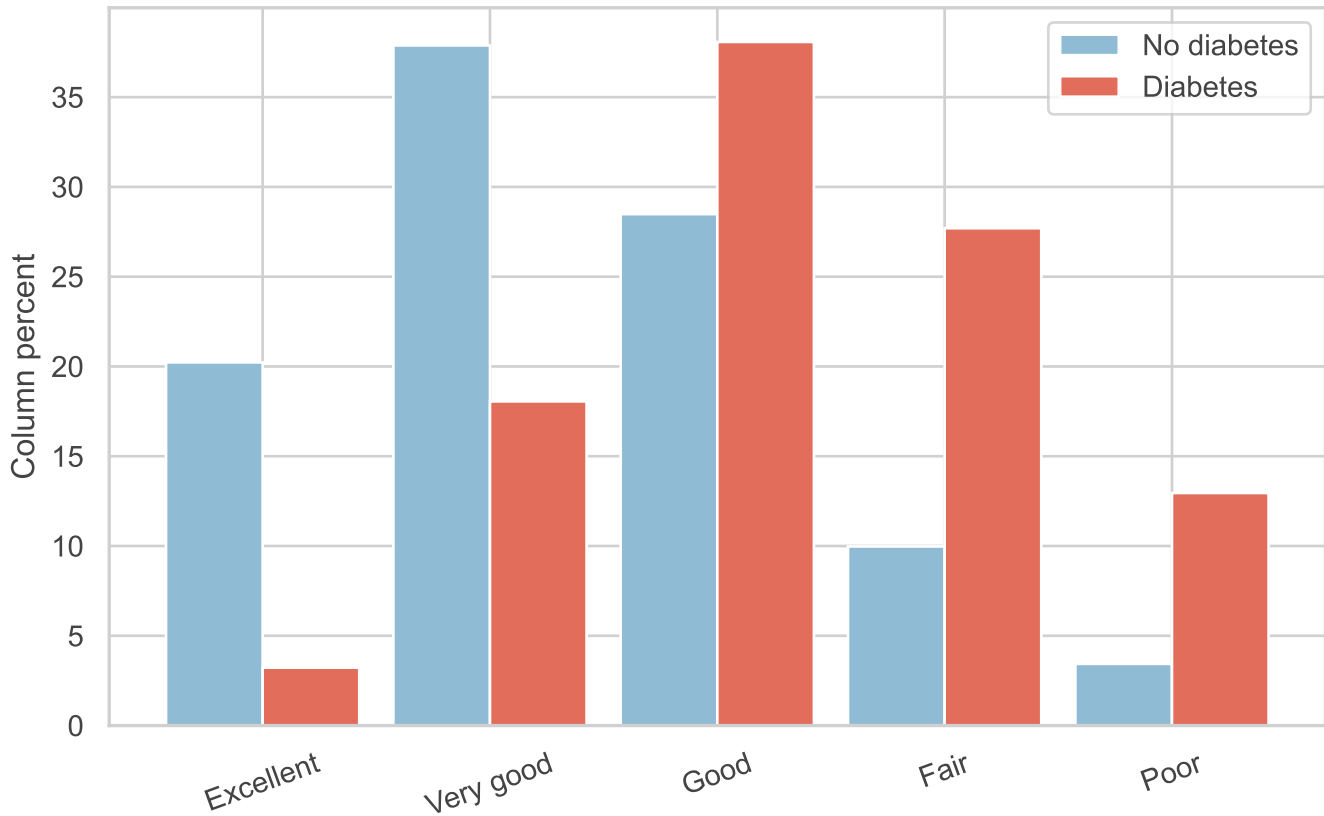
General Health Baseline Comparison by Diabetes Status

Variable	No diabetes	Diabetes	Difference	SMD
GenHlth	2.385	3.291	0.906	0.891
PhysHlth	3.641	7.954	4.313	0.439
MentHlth	2.978	4.462	1.484	0.184
DiffWalk	0.135	0.371	0.236	0.563
PhysHlthPct30	12.137	26.515	14.378	0.439
MentHlthPct30	9.927	14.873	4.946	0.184

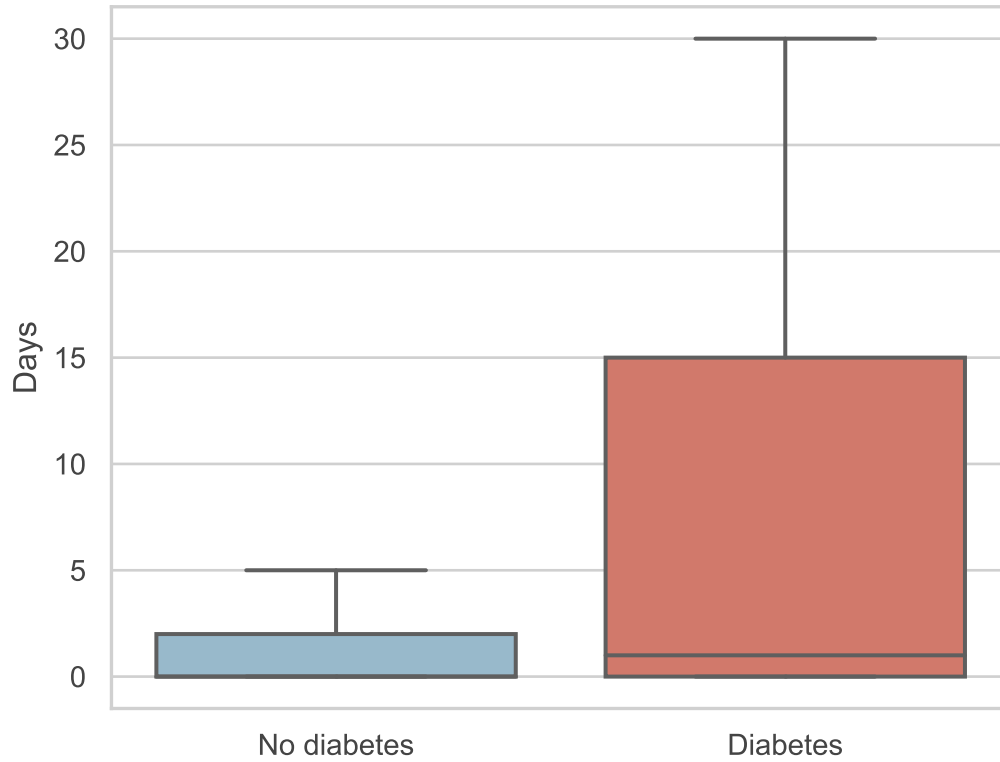
General Health Diabetes Prevalence by Category

Variable	Category	Count	Diabetes prevalence
GenHlth	Excellent	45,299	2.5%
GenHlth	Very good	89,084	7.2%
GenHlth	Good	75,646	17.8%
GenHlth	Fair	31,570	31.0%
GenHlth	Poor	12,081	37.9%
DiffWalk	No	211,005	10.5%
DiffWalk	Yes	42,675	30.7%
PhysHlth	0	160,052	10.5%
PhysHlth	1-5	46,811	14.1%
PhysHlth	6-13	13,157	20.3%
PhysHlth	14-29	14,260	25.5%
PhysHlth	30	19,400	29.5%
MentHlth	0	175,680	13.3%
MentHlth	1-5	41,792	11.9%
MentHlth	6-13	11,671	15.7%
MentHlth	14-29	12,449	19.2%
MentHlth	30	12,088	22.9%

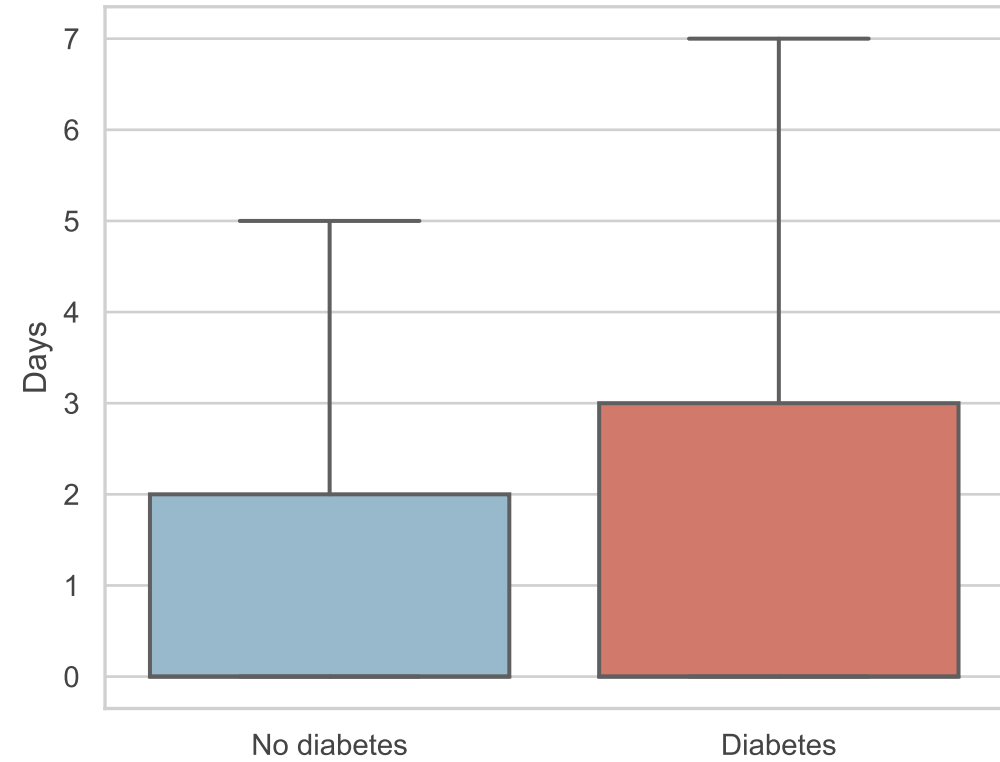
General Health Distribution by Diabetes Status



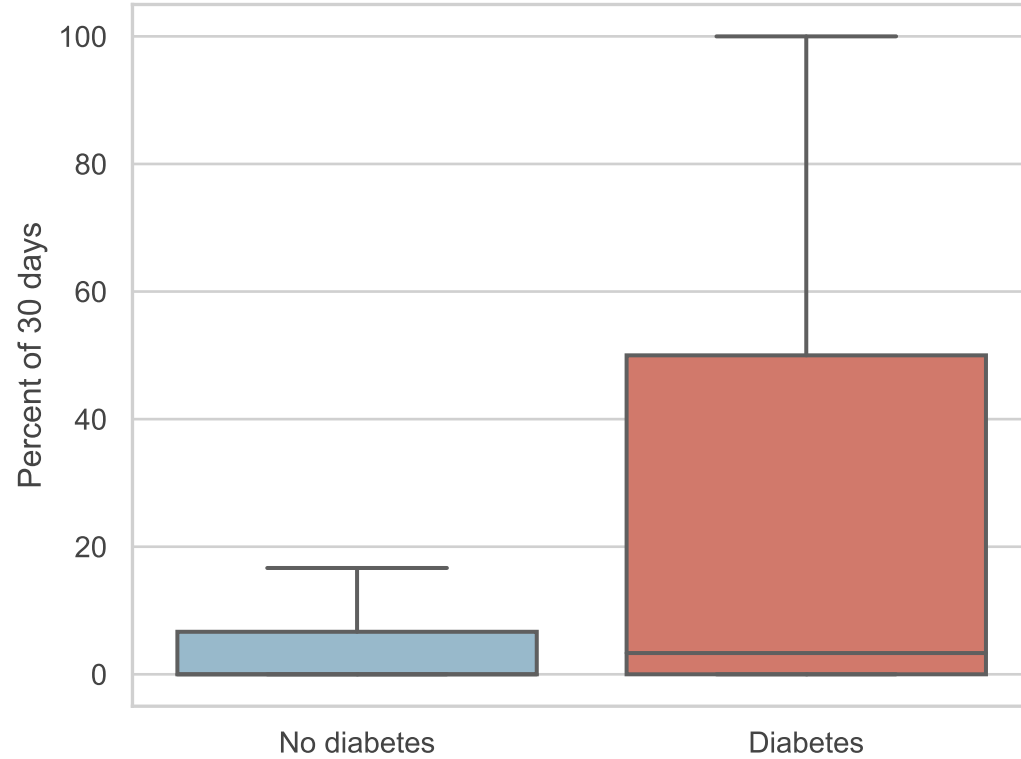
Poor Physical Health Days



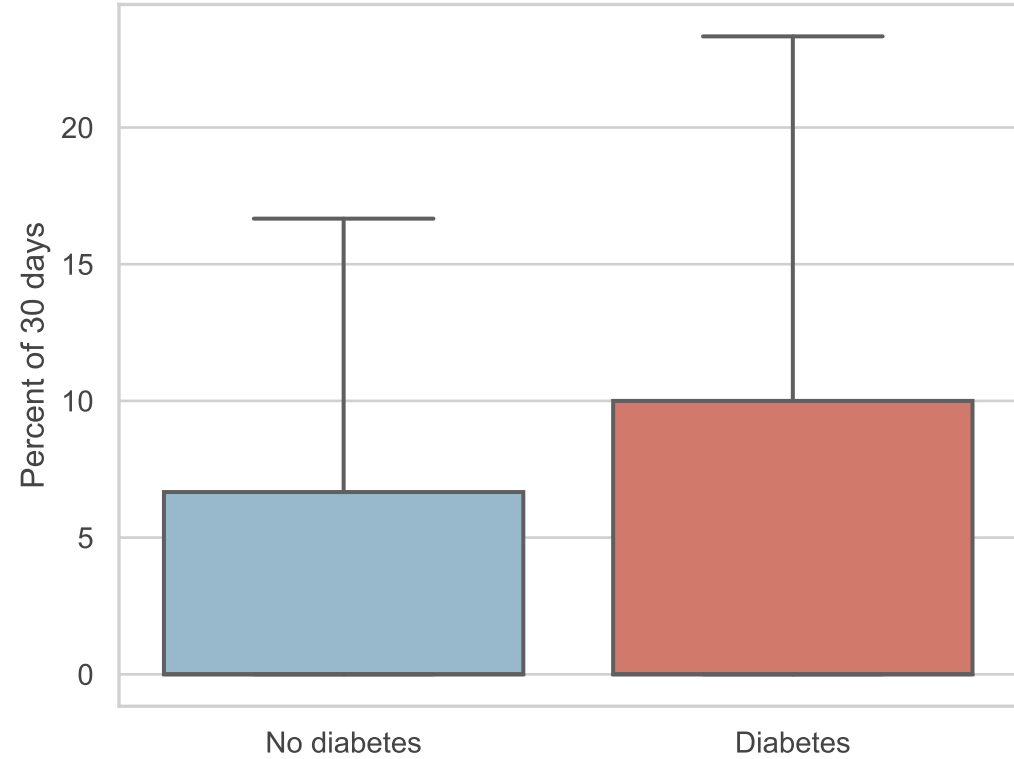
Poor Mental Health Days



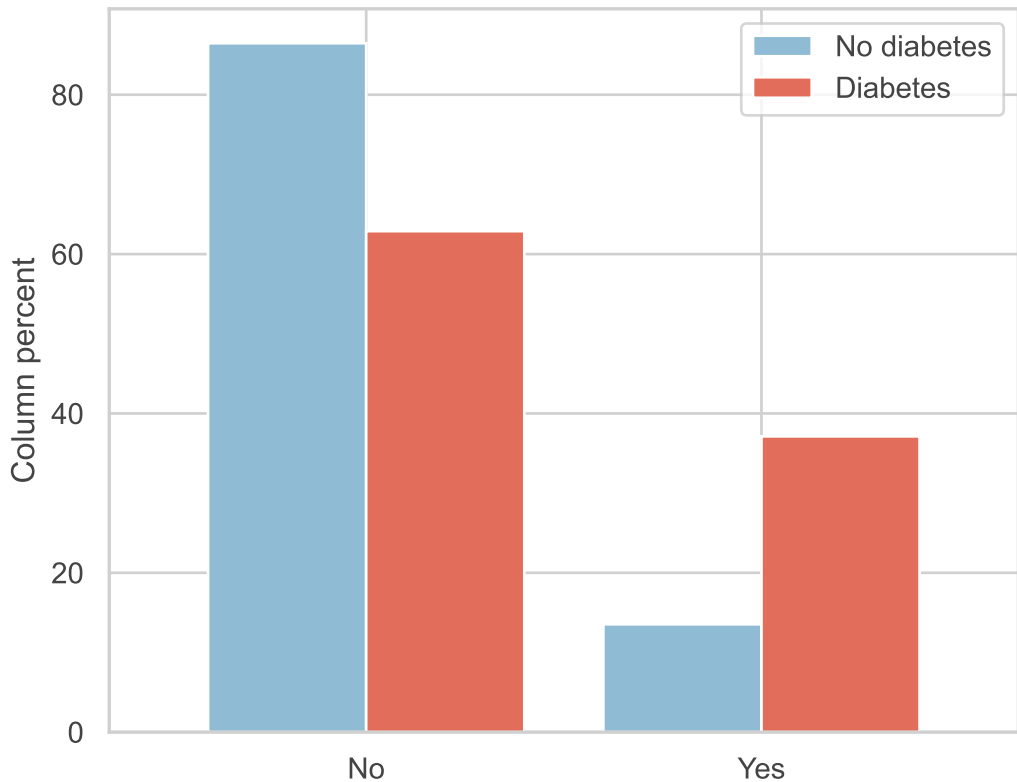
Poor Physical Health as % of 30 Days



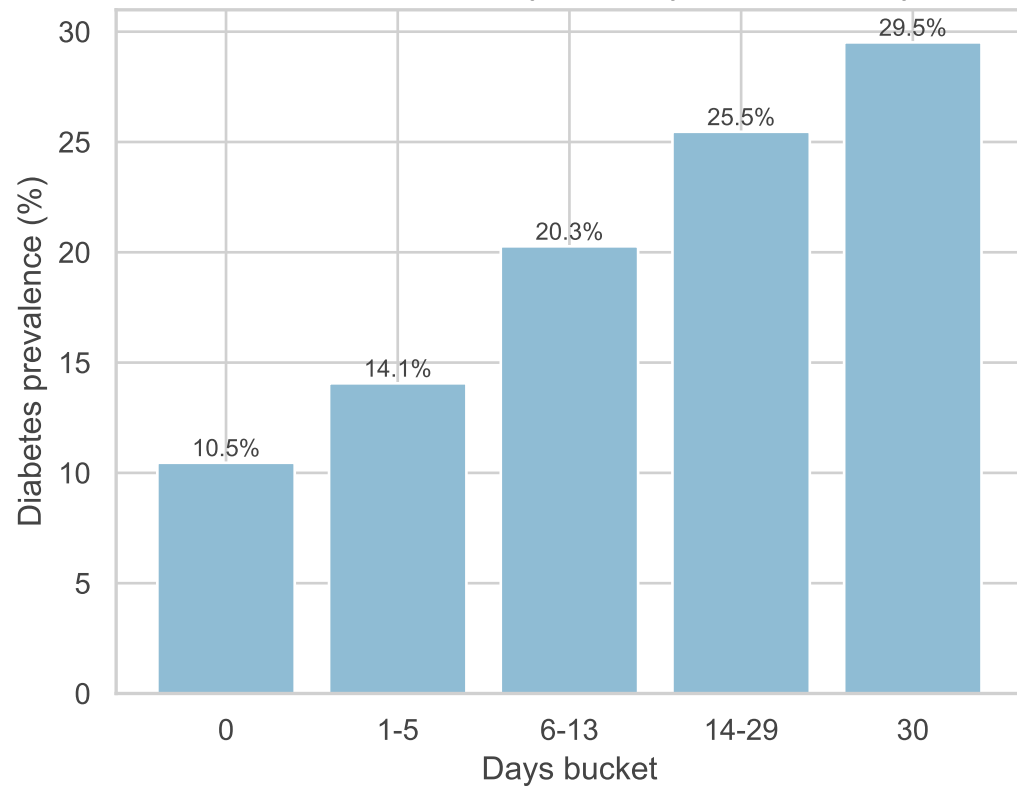
Poor Mental Health as % of 30 Days



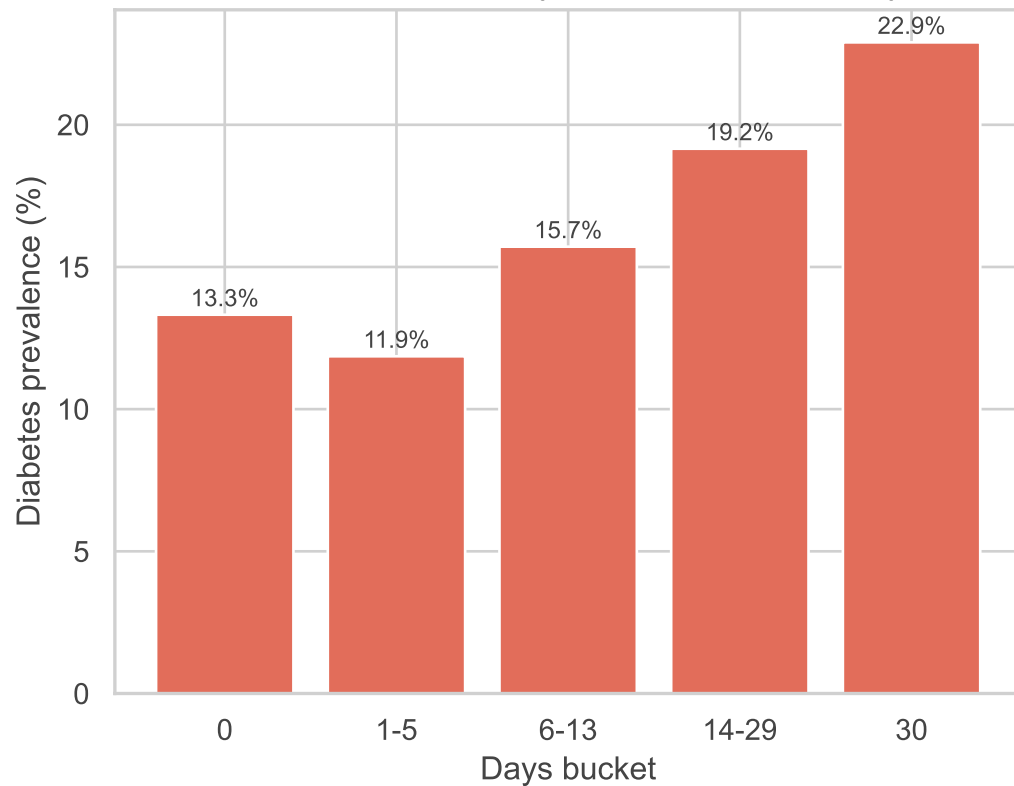
Walking Difficulty Distribution by Diabetes Status



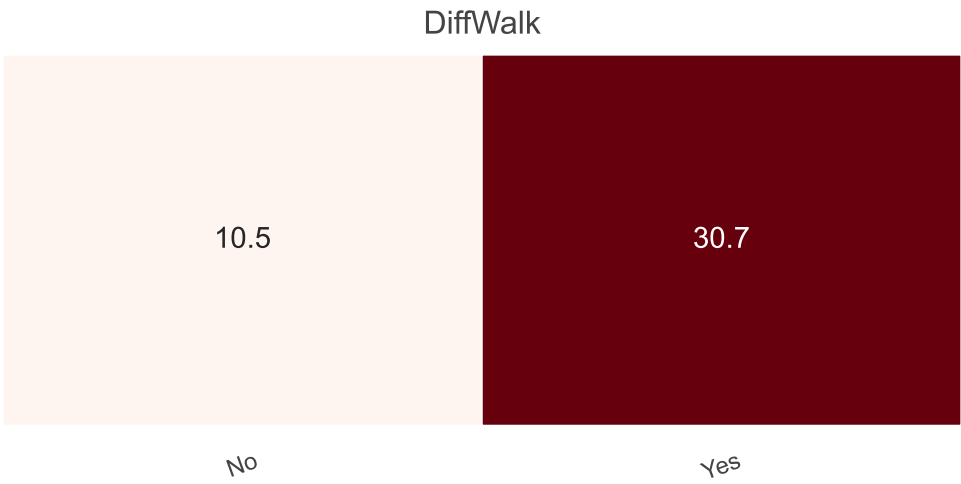
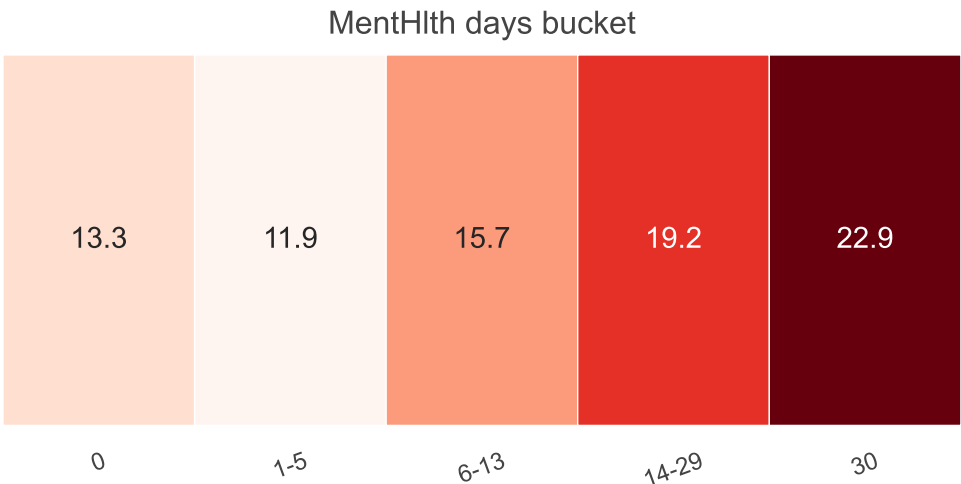
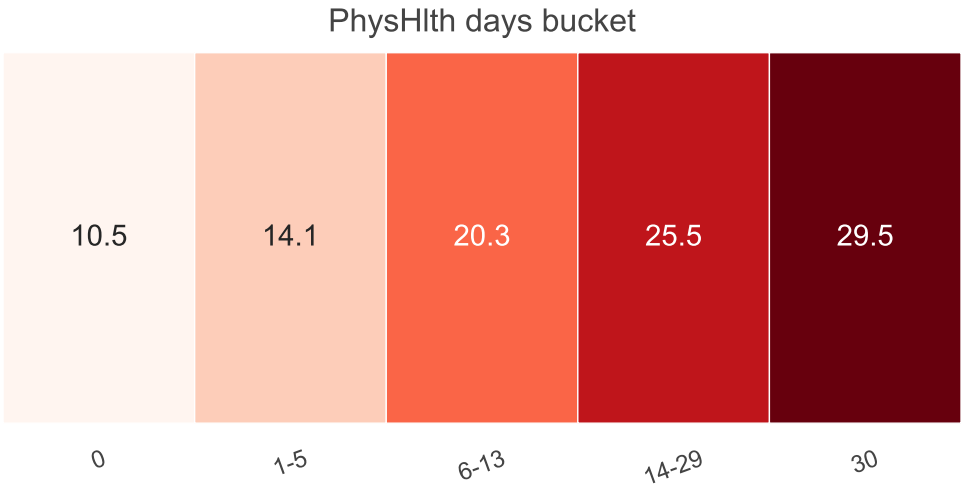
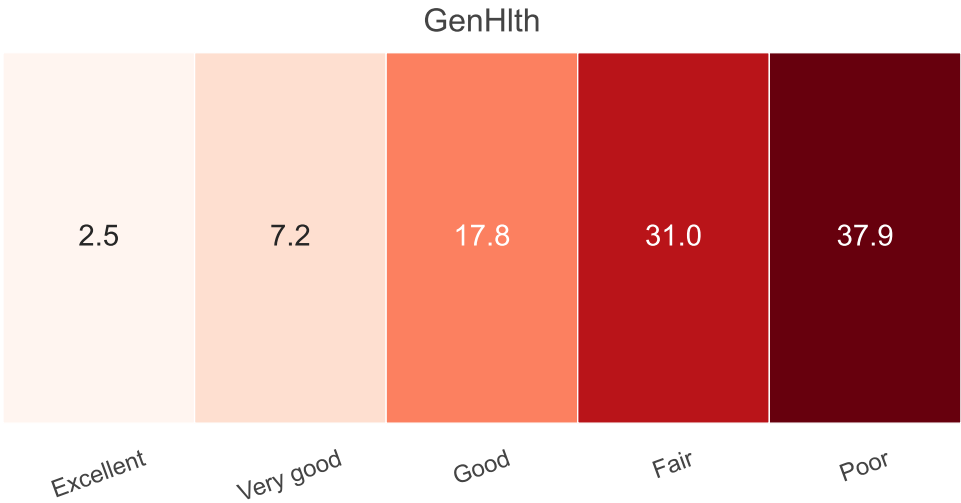
Diabetes Prevalence by Poor Physical Health Days



Diabetes Prevalence by Poor Mental Health Days



General Health Prevalence Heatmaps



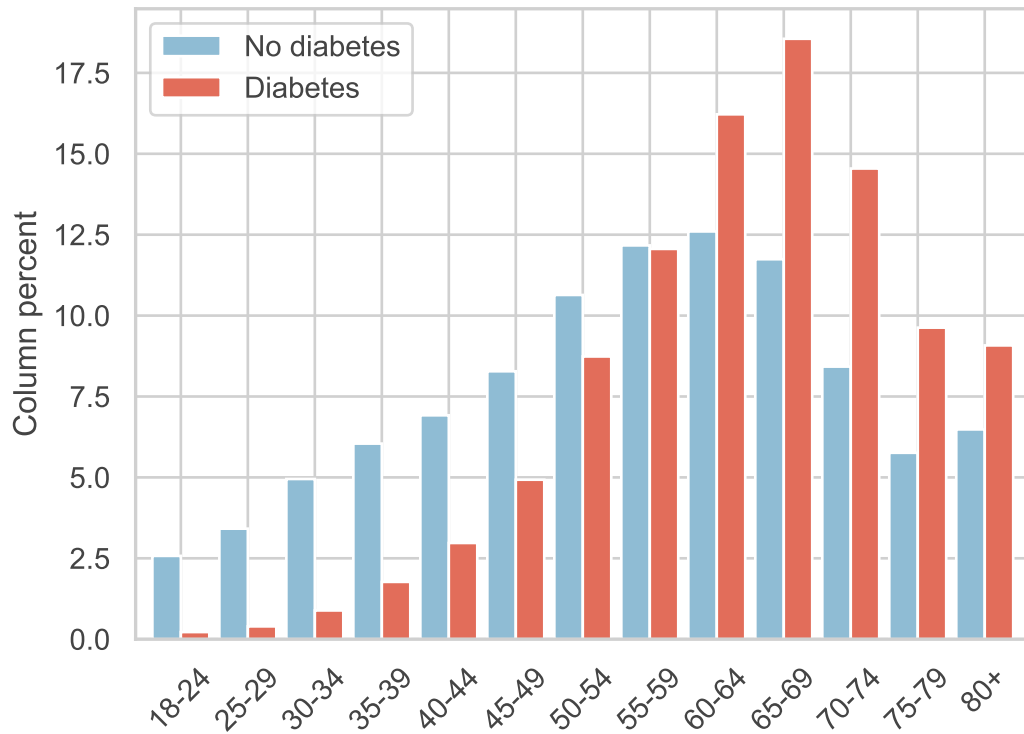
Demographic / Fairness Baseline Comparison by Diabetes Status

Variable	No diabetes	Diabetes	Difference	SMD
Age	7.814	9.379	1.565	0.571
Sex	0.434	0.479	0.045	0.091
Income	6.19	5.21	-0.98	-0.463
Education	5.1	4.746	-0.354	-0.35

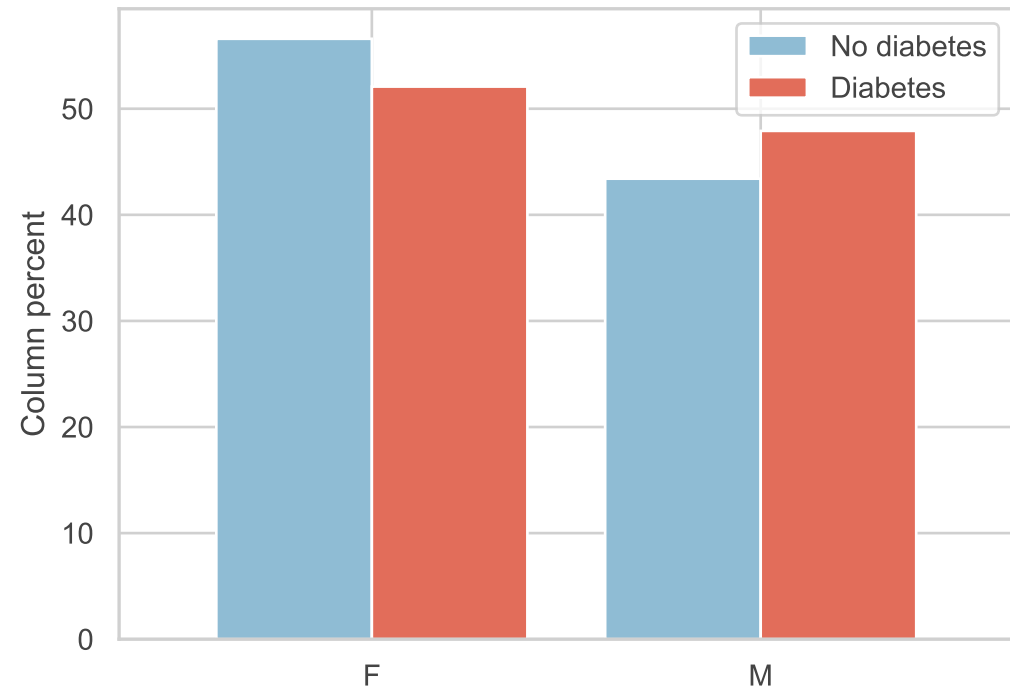
Demographic / Fairness Diabetes Prevalence by Category

Variable	Category	Count	Diabetes prevalence
Age	18-24	5,700	1.4%
Age	25-29	7,598	1.8%
Age	30-34	11,123	2.8%
Age	35-39	13,823	4.5%
Age	40-44	16,157	6.5%
Age	45-49	19,819	8.8%
Age	50-54	26,314	11.7%
Age	55-59	30,832	13.8%
Age	60-64	33,244	17.2%
Age	65-69	32,194	20.4%
Age	70-74	23,533	21.8%
Age	75-79	15,980	21.3%
Age	80+	17,363	18.5%
Sex	F	141,974	13.0%
Sex	M	111,706	15.2%
Income	<10k	9,811	24.3%
Income	10-15k	11,783	26.2%
Income	15-20k	15,994	22.3%
Income	20-25k	20,135	20.1%
Income	25-35k	25,883	17.4%
Income	35-50k	36,470	14.5%
Income	50-75k	43,219	12.2%
Income	75k+	90,385	8.0%
Education	No school/K	174	27.0%
Education	Grades 1-8	4,043	29.3%
Education	Grades 9-11	9,478	24.2%
Education	HS graduate	62,750	17.6%
Education	Some college	69,910	14.8%
Education	College grad	107,325	9.7%

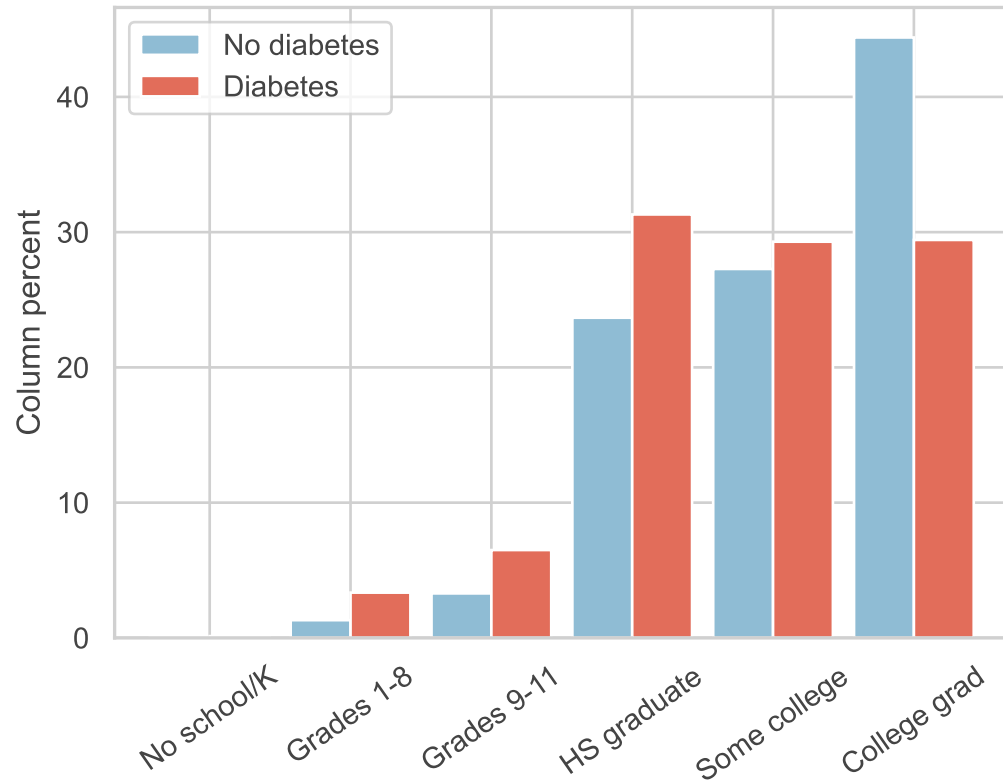
Age Distribution by Diabetes Status



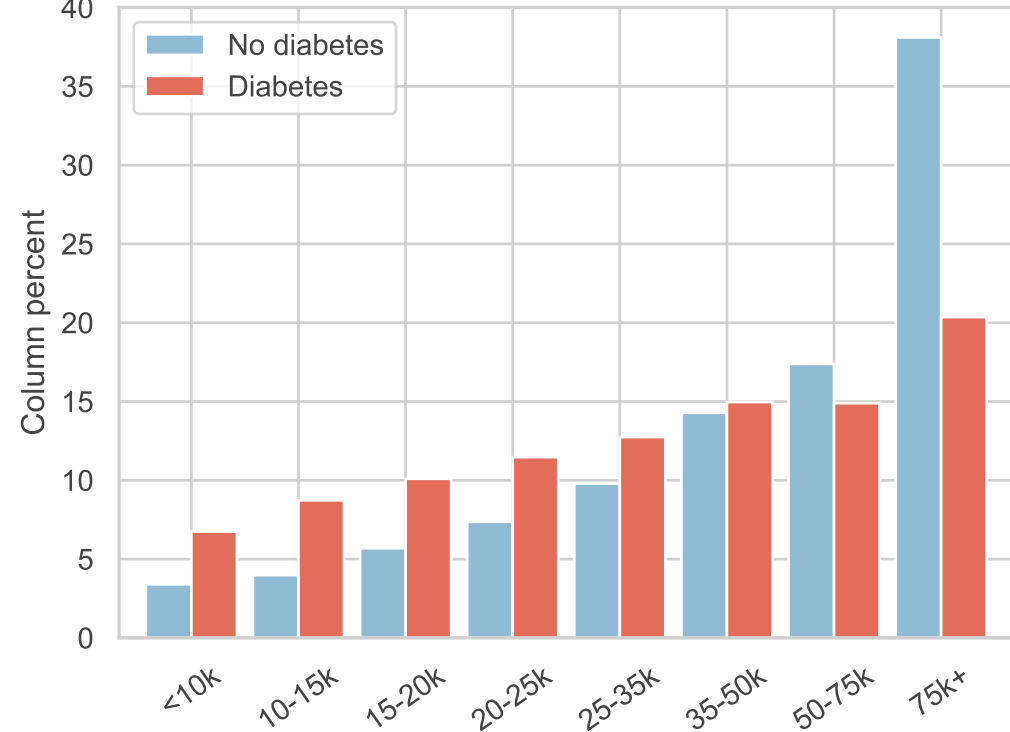
Sex Distribution by Diabetes Status



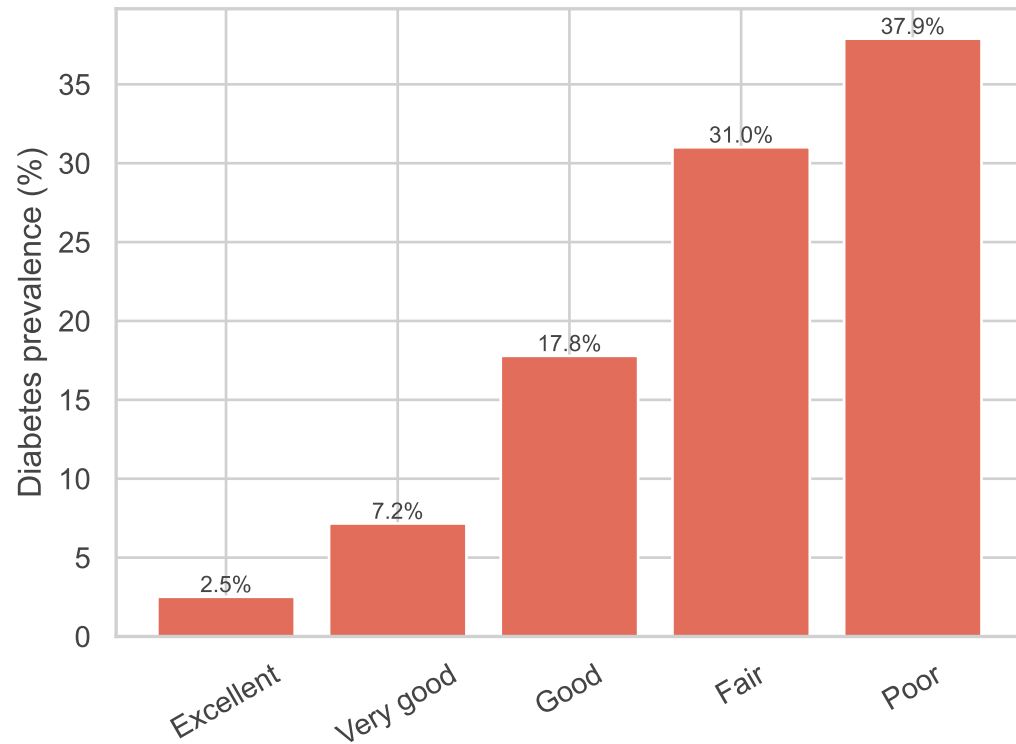
Education Distribution by Diabetes Status



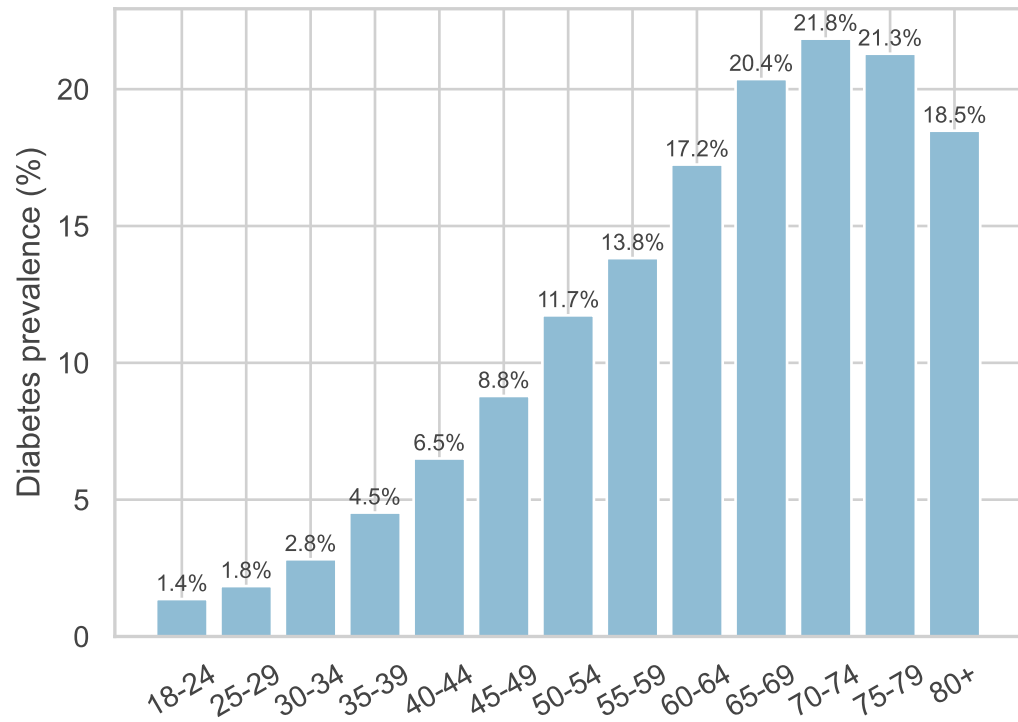
Income Distribution by Diabetes Status



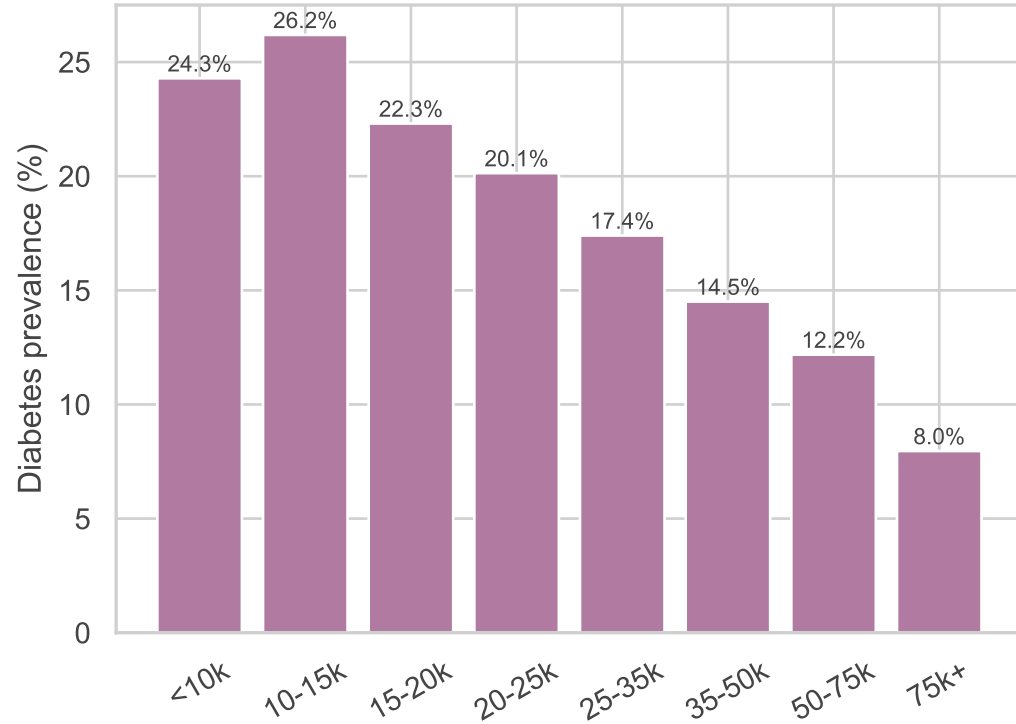
Prevalence by Self-Rated Health



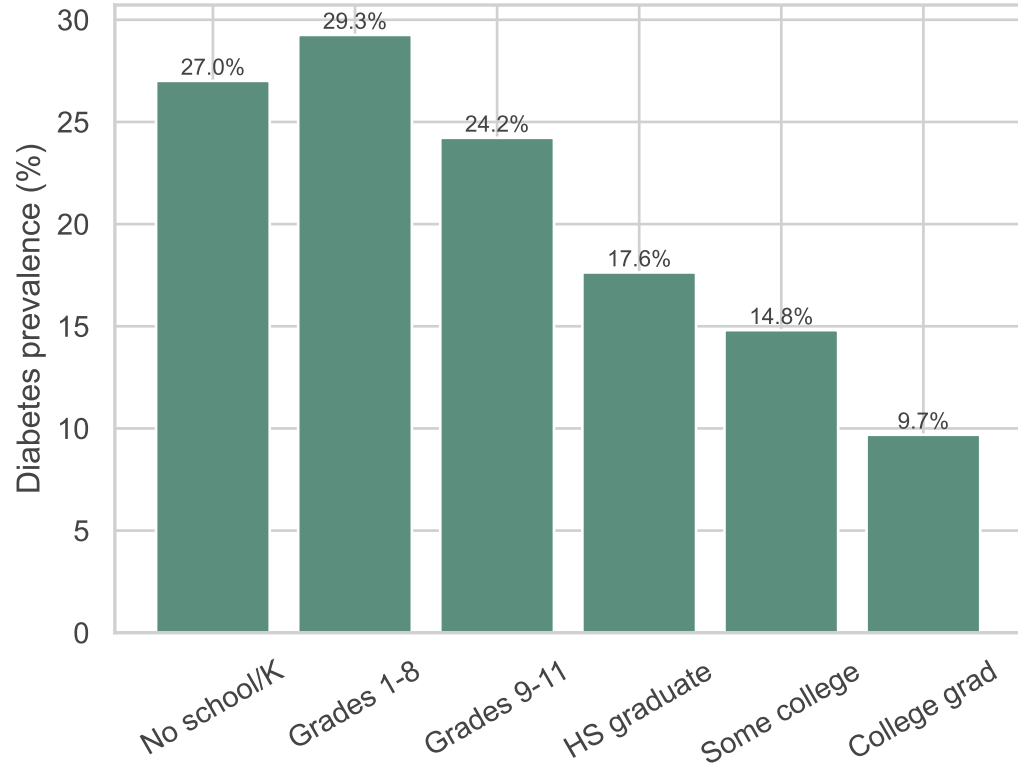
Prevalence by Age Group



Prevalence by Income Category

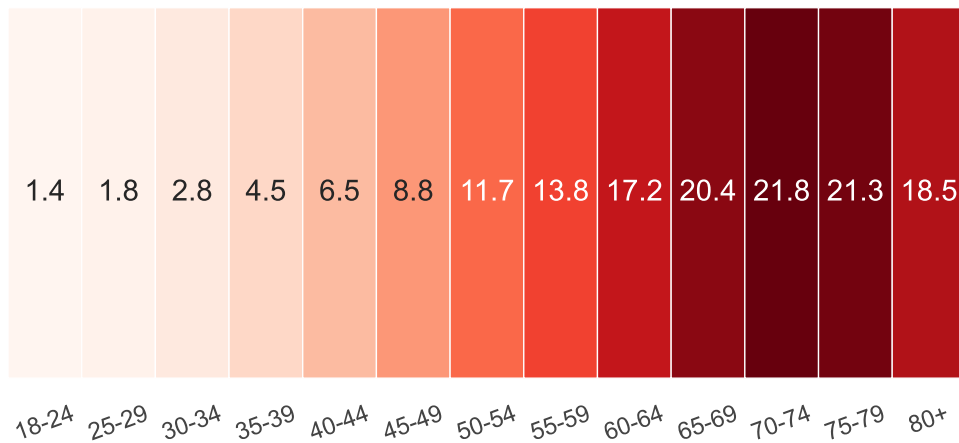


Prevalence by Education Level

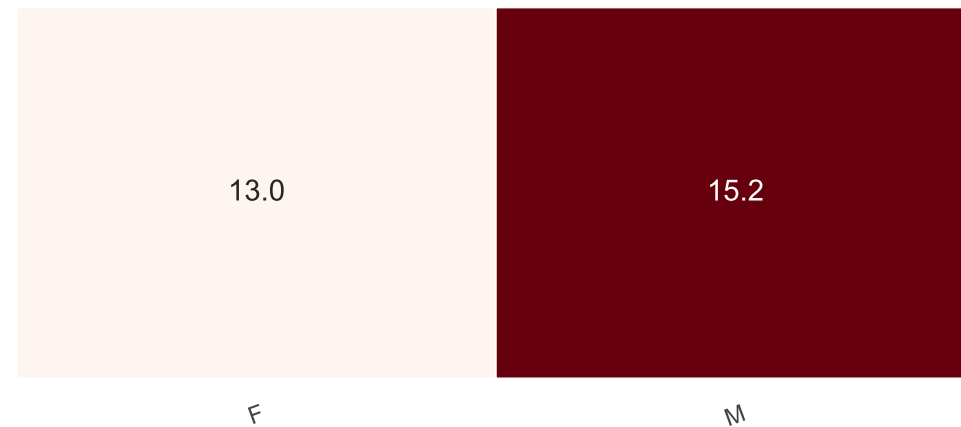


Demographic / Fairness Prevalence Heatmaps

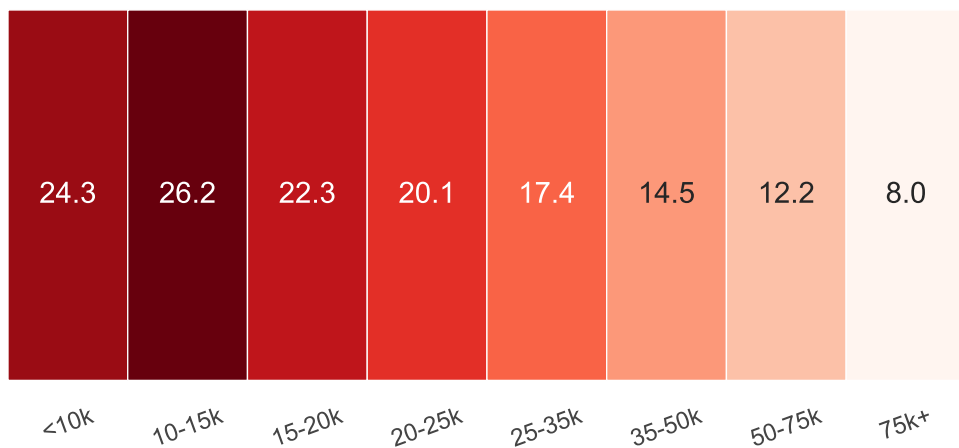
Age



Sex



Income



Education

